PROJECT COLLECTION

REZA BABAPIRI

Control Systems & Automation

Feedback control, power electronics and estimation

4 TECHNICAL CASE STUDIES

Engineering scope / Method / Results / Visual evidence / Limitations

Technical design, coding and analysis: Reza Babapiri

reza-babapiri.github.io

POWER ELECTRONICS · SIMULINK

Buck Converter | Cascaded PI Control

100 V to 48 V, 1 kW converter design with current and voltage feedback loops

ENGINEERING QUESTION

How can a 100 V-to-48 V converter maintain output regulation under load changes without excessive ripple or poorly damped transients?

ROLE & CONTRIBUTION

I prepared the converter sizing calculations, single-loop and cascaded PI-controller designs, Simulink implementations and comparative evaluation.

ENGINEERING DECISIONS

- Calculated nominal LC values from ripple design constraints, then iterated passive-component values using simulated load and step responses.
- Implemented an inner inductor-current PI loop and an outer output-voltage PI loop; retained the single-loop model as a point of comparison.
- Examined full-load and no-load Bode plots, controller phase margins, steady-state ripple and startup transient behavior.
- Distinguished between outcomes that meet nominal specifications and the problematic lightly loaded response.

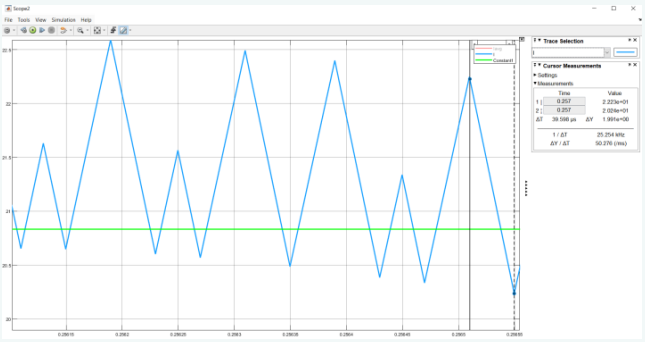
SELECTED ENGINEERING EVIDENCE

Simulated nominal operating point: 100 V input, 48 V output, 1 kW, 50 kHz; reported voltage ripple ~1.1%, current ripple ~9.6%, full-load output overshoot 1.69%. No-load transient overshoot reached ~75%.

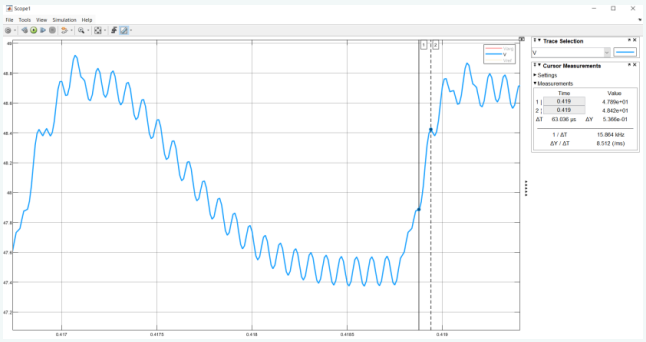
VISUAL EVIDENCE · POWER ELECTRONICS · SIMULINK

Buck Converter | Cascaded PI Control

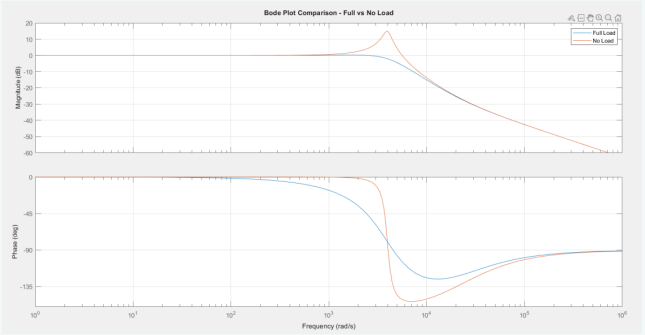
Original images, models and results from the supplied project material



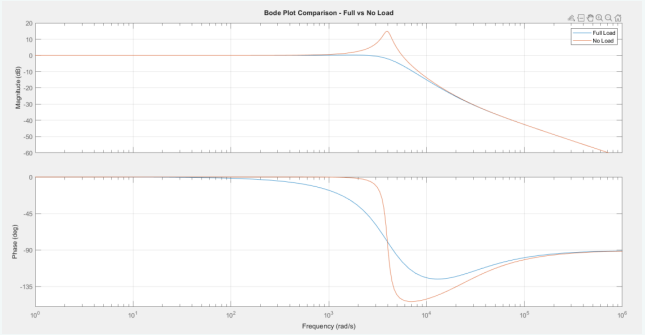
Inductor / output-current trace and steady-state ripple measurement from the simulation.



Voltage-related steady-state trace from the original project report.



Frequency-domain comparison across loading conditions; gain and phase response.



Full-load versus no-load Bode response taken from the report.

RESULTS & ENGINEERING INTERPRETATION

Simulated nominal operating point: 100 V input, 48 V output, 1 kW, 50 kHz; reported voltage ripple ~1.1%, current ripple ~9.6%, full-load output overshoot 1.69%. No-load transient overshoot reached ~75%.

STATE-SPACE CONTROL · NONLINEAR SIMULATION

Inverted Pendulum | LQR & LQI

Cart–pendulum stabilization, tracking, saturation and anti-windup

ENGINEERING QUESTION

How should a controller stabilize an inherently unstable cart-pendulum equilibrium while respecting finite actuation?

ROLE & CONTRIBUTION

I modelled and analysed the cart-pendulum dynamics, implemented controller comparisons, and evaluated tracking and saturation behavior in MATLAB/Simulink.

ENGINEERING DECISIONS

- Linearized the system about the upright equilibrium and examined controllability and unstable pole locations.
- Compared pole placement with optimal LQR state feedback, then introduced integral action for reference tracking.
- Investigated nonlinear simulation, bounded control input, saturation and anti-windup implications.

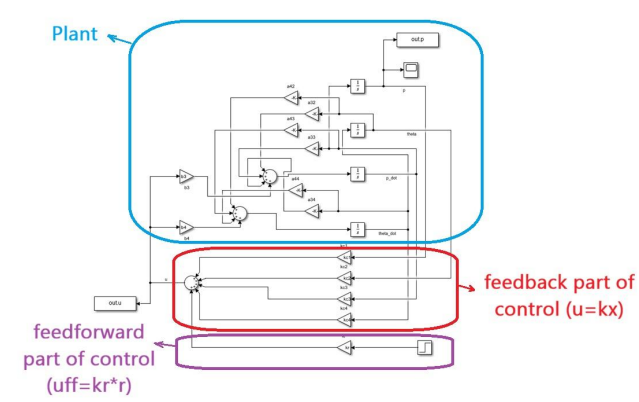
SELECTED ENGINEERING EVIDENCE

Controller-response plots and MATLAB/Simulink scripts show the comparative simulated dynamics and trade-offs. No physical cart-pendulum rig was tested.

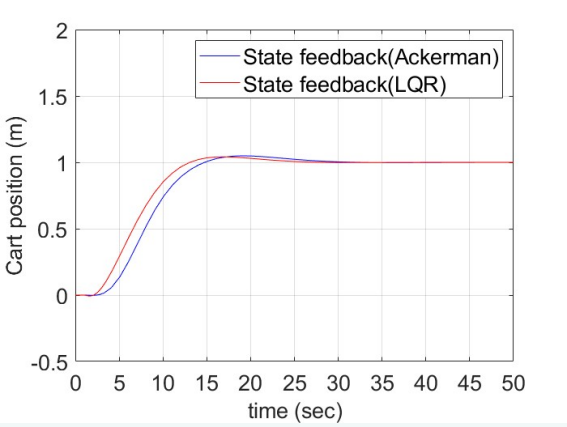
VISUAL EVIDENCE · STATE-SPACE CONTROL · NONLINEAR SIMULATION

Inverted Pendulum | LQR & LQI

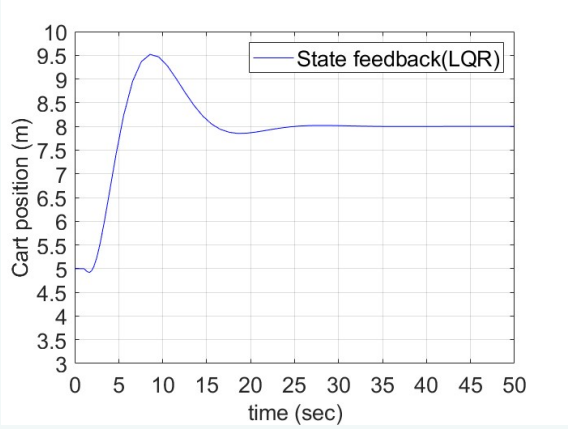
Original images, models and results from the supplied project material



Controller and response comparison from the cart-pendulum analysis.



Model response related to state feedback and transient behavior.



Additional stability / tracking plot from the course report.

RESULTS & ENGINEERING INTERPRETATION

Controller-response plots and MATLAB/Simulink scripts show the comparative simulated dynamics and trade-offs. No physical cart-pendulum rig was tested.

CONTROL ENGINEERING · ROBUSTNESS

Vehicle Cruise Control | PI & Anti-Windup

Speed tracking under hills, actuator saturation and parameter changes

ENGINEERING QUESTION

How can a longitudinal vehicle-speed controller handle road grades and saturated throttle commands without excessive windup?

ROLE & CONTRIBUTION

I wrote the modelling and control simulations, studied disturbances and anti-windup and analysed sensitivity/robustness in coursework.

ENGINEERING DECISIONS

- Built a simplified longitudinal vehicle model and tuned PI speed regulation.
- Introduced gradient disturbances and actuator saturation to challenge the nominal design.
- Compared the transient under anti-windup and unmodified integral action; inspected sensitivity transfer functions.

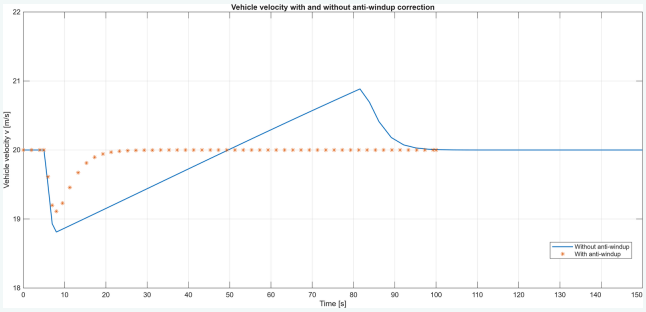
SELECTED ENGINEERING EVIDENCE

One reported disturbance scenario shows a peak speed overshoot near 0.02 m/s with anti-windup versus ~0.885 m/s without anti-windup. These figures are specific to that setup.

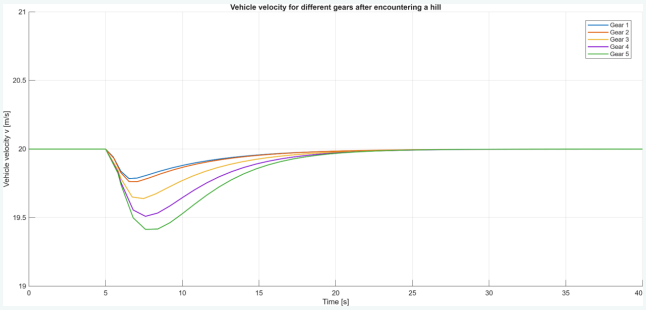
VISUAL EVIDENCE · CONTROL ENGINEERING · ROBUSTNESS

Vehicle Cruise Control | PI & Anti-Windup

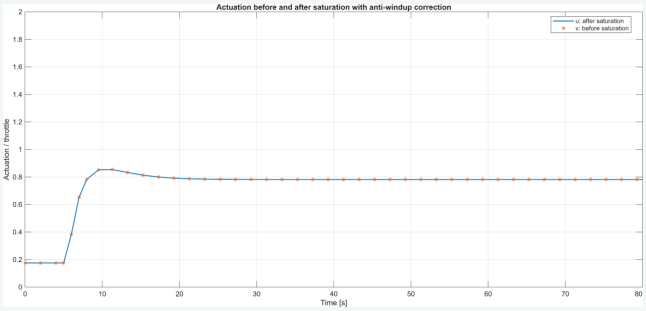
Original images, models and results from the supplied project material



Course report vehicle-speed control response under defined simulation conditions.



PI cruise-control transient and disturbance-response comparison.



Additional robustness and anti-windup results from coursework.

RESULTS & ENGINEERING INTERPRETATION

One reported disturbance scenario shows a peak speed overshoot near 0.02 m/s with anti-windup versus ~0.885 m/s without anti-windup. These figures are specific to that setup.

ESTIMATION · STOCHASTIC SYSTEMS

Kalman Filtering | Penalty-Kick Tracking

Tracking a moving target from noisy 2-D measurements

ENGINEERING QUESTION

How can a noisy moving-target trajectory be estimated using a simple state-space motion model?

ROLE & CONTRIBUTION

I implemented the simulated penalty-kick tracking model, Kalman estimator, numerical experiments and plotted evaluation.

ENGINEERING DECISIONS

- Selected a constant-velocity state representation with explicit process and observation uncertainty.
- Propagated state predictions and corrected them with noisy measured positions.
- Compared estimated and reference trajectories across ten simulated initial-condition variants.

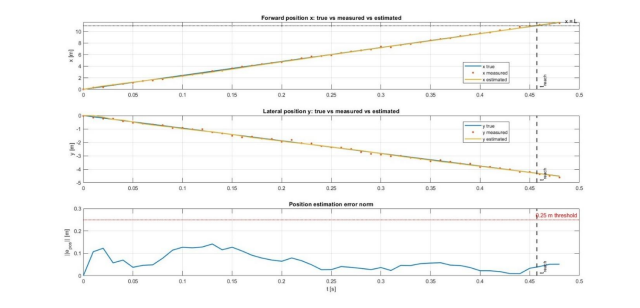
SELECTED ENGINEERING EVIDENCE

The coursework report states that all ten simulated trials satisfied a 0.25 m endpoint tracking-error requirement.

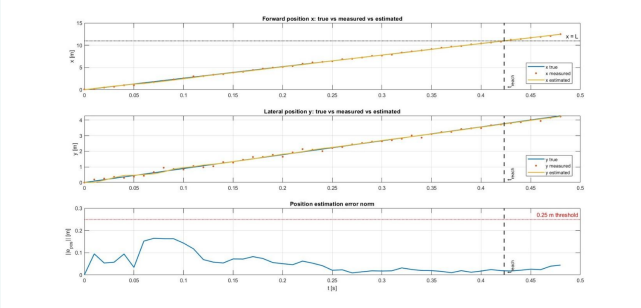
VISUAL EVIDENCE · ESTIMATION · STOCHASTIC SYSTEMS

Kalman Filtering | Penalty-Kick Tracking

Original images, models and results from the supplied project material



Kalman-filter estimate alongside simulated noisy target trajectory observations.



Estimation tracking comparison from the simulated penalty kick.

RESULTS & ENGINEERING INTERPRETATION

The coursework report states that all ten simulated trials satisfied a 0.25 m endpoint tracking-error requirement.